Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2006-0412-DFR-2	Issued	2006-02-23

Daily Field Report 2 - UT Thickness Survey - ATU-100 Full Unit

Scope and Responsibilities

Kevin Doucette, acting as lead UT technician, performed ultrasonic thickness measurements on specified piping within the ATU-100. The scope of work was executed under the authority of QSCP-WP-2006-02226.

The survey was conducted in accordance with the site NDE procedure QSCP-NDT-001 Rev 0 and the Halward Inspection Services Ltd. procedure HIS-UT-002 Rev 1. All measurements were performed following the guidelines of ASTM E797.

Equipment and Calibration

Thickness measurements were obtained using a Krautkramer DMS 2 0317. The instrument's calibration status is verified by HIS-CAL-06-012.

Field calibration and verification were performed using the HIS-STB-02 calibration block to ensure accuracy and repeatability of the readings.

Inspection Results - Rich Amine Service

Thickness readings were taken on line 4-RA-102 (A106 Gr.B, OD 114.3 mm, t_nom 8.56 mm). The design limits for this line are 862 kPa(g) and 82 °C, with a t-min of 3.81 mm.

Measurements were also recorded for the rich amine header 6-RA-103 (A106 Gr.B, OD 168.3 mm, t_nom 10.97 mm). This line has a t-min of 4.57 mm and design limits of 862 kPa(g) and 82 °C.

Inspection Results - Regenerator Overhead

The regenerator overhead line 3-RV-101 (A106 Gr.B, OD 88.9 mm, t_nom 7.62 mm) was surveyed. Design parameters are 345 kPa(g) and 121 °C, with a t-min of 3.56 mm.

The regenerator overhead line 2-RV-102 (A106 Gr.B, OD 60.3 mm, t_nom 5.54 mm) was surveyed. Design parameters are 345 kPa(g) and 121 °C, with a t-min of 2.79 mm.

Inspection Results - Reflux Service

Thickness measurements were performed on the reflux line 3-RF-101 (A312 TP316, OD 88.9 mm, t_nom 5.49 mm).

The design limits for 3-RF-101 are 517 kPa(g) and 104 °C, with a retirement thickness (t-min) of 2.41 mm.

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-23	_____